

Kink dynamics for the Yang-Mills field in an extremal Reissner-Nordström black hole

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Some previous results for **Yang-Mills fields**.

1. The global dynamics when it propagates in a 4-D **Minkowski spacetime** is well-understood: All solutions starting from smooth initial data remain smooth for all times and **decay to zero** as time tends to infinity ([Eardley-Moncrief 81](#), [Christodoulou 81](#), [Bizoń-Chmaj-Rostworowski 07](#)).
2. Existence of **nontrivial stationary solutions** which play the role of unstable **attractors**, as was shown for the Schwarzschild background ([Bizoń-Rostworowski-Zenginoglu 10](#)).
3. Existence of infinitely many static solutions on the **exterior of the extremal Reissner-Nordström black hole** ([Bizoń-Kahl 16](#)).

The exterior of the *extremal* ($Q = M$) Reissner-Nordström black hole is a globally hyperbolic static spacetime $(\mathcal{M}, \hat{g})$ where

$$\hat{g} := - \left(1 - \frac{M}{r}\right)^2 dt^2 + \left(1 - \frac{M}{r}\right)^{-2} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2),$$

with $t \in \mathbb{R}$, $r > M$, $(\theta, \phi) \in \mathbb{S}^2$, and $M > 0$ a positive constant. Applying

$$\tau = \frac{t}{4M} \in \mathbb{R}, \quad x = \log \left(\frac{r}{M} - 1 \right) \in \mathbb{R}, \quad \hat{g} = 16M^2(1 + e^{-x})^{-2}g,$$

then a geodesically complete spacetime $(\mathcal{M}, g)$ is obtained, where

$$g = -d\tau^2 + \cosh^4 \left(\frac{x}{2} \right) (dx^2 + d\theta^2 + \sin^2 \theta d\phi^2).$$

BIZOŃ-KAHL SETTING

In terms of the Yang-Mills field

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu + [A_\mu, A_\nu],$$

the lagrangian density reads

$$\mathcal{L} = \text{Tr}(F_{\alpha\beta} F_{\mu\nu} g^{\alpha\mu} g^{\beta\nu}) \sqrt{-\det g_{\mu\nu}}.$$

$$\begin{array}{ccc} \text{Conformal structure} & + & \text{Conformal invariance} \\ \text{extr. RN BH} & & \text{YM eqs} \end{array} \implies \text{Non-trivial Solutions}$$

Bizoń and Kahl proposed the spherically symmetric, purely magnetic ansatz

$$A = \varphi(t, x)(\tau_1 d\theta + \tau_2 \sin\theta d\phi) + \tau_3 \cos\theta d\phi,$$

where $\varphi = \varphi(t, x)$ is a real scalar field.

This gives the reduced, variable coefficients Lagrangian density

$$\mathcal{L} = -\frac{1}{2} \cosh^2\left(\frac{x}{2}\right) (\partial_t \varphi)^2 + \frac{1}{2} \operatorname{sech}^2\left(\frac{x}{2}\right) \left((\partial_x \varphi)^2 + \frac{1}{2} (1 - \varphi^2)^2 \right).$$

The associated Euler-Lagrange equation for the field φ , up to scaling $t \mapsto \frac{3}{2}t$, equivalent to the associated Yang-Mills model, is given by

$$\partial_t^2 \varphi - Q \partial_x (Q \partial_x \varphi) + Q^2 (\varphi^2 - 1) \varphi = 0, \quad (1)$$

with $Q(x) := \frac{3}{2} \operatorname{sech}^2\left(\frac{x}{2}\right)$ the standard KdV soliton, and **energy** given by

$$\mathcal{E}[\varphi, \varphi_t] = \int \left(\frac{1}{2} Q^{-1} (\partial_t \varphi)^2 + \frac{1}{2} Q \left((\partial_x \varphi)^2 + \frac{1}{2} (1 - \varphi^2)^2 \right) \right) dx$$

formally conserved along the flow.

The first non-trivial static solutions of (1) are given by

$$H(x) := H_1(x) = \tanh\left(\frac{x}{2}\right), \quad H_2(x) = \frac{2 \cosh x - 2 - \sqrt{6}}{2 \cosh x + 4 + 3\sqrt{6}}.$$

Following Bizoń and Kahl, we introduce the change of variables

$$\alpha(x) = \frac{1}{3}(\sinh x + x),$$

strictly monotone and bijective from $\mathbb{R}$ onto itself, with α^{-1} not having an exact form and with logarithmic growth.

We define the **distorted soliton** and **kink**:

$$\tilde{Q}(x) = Q(\alpha^{-1}(x)), \quad \tilde{H}(x) = H(\alpha^{-1}(x)),$$

Both functions have only a polynomial rate of convergence at infinity, with

$$0 \leq \tilde{Q}(x) \lesssim \frac{1}{|x|}, \quad \left| \tilde{H}(x) \mp 1 \right| \lesssim \frac{1}{|x|}, \quad \text{as } x \rightarrow \pm\infty.$$

ENERGY SPACE

With this, ϕ is a solution of the original equation iff $\phi = \phi \circ \alpha^{-1}$ solves

$$\partial_t^2 \phi - \partial_x^2 \phi - \tilde{Q}^2(\phi - \phi^3) = 0,$$

with **kink solution** $\tilde{H}(x)$. Let $\phi = (\phi, \partial_t \phi) = (\phi_1, \phi_2)$, then

$$\begin{cases} \partial_t \phi_1 = \phi_2 \\ \partial_t \phi_2 = \partial_x^2 \phi_1 + \tilde{Q}^2(x)(1 - \phi_1^2)\phi_1. \end{cases} \quad (2)$$

The conserved energy reads now

$$E[\phi_1, \phi_2] = \frac{1}{2} \int \left(\phi_2^2 + (\partial_x \phi_1)^2 + \frac{1}{2} \tilde{Q}^2(1 - \phi_1^2)^2 \right) dx.$$

Thus, the energy makes sense for the set of functions

$$\mathbf{E} = \left\{ (\phi_1, \phi_2) \in (L_{\text{loc}}^1(\mathbb{R}))^2 : \partial_x \phi_1, \tilde{Q}(1 - \phi_1^2), \phi_2 \in L^2(\mathbb{R}) \right\}.$$

We endow it with the norm

$$\|\mathbf{m}\|_{(H_0 \times L^2)(\mathbb{R})}^2 := \|\partial_x m_1\|_{L^2(\mathbb{R})}^2 + \|\tilde{Q} m_1\|_{L^2(\mathbb{R})}^2 + \|m_2\|_{L^2(\mathbb{R})}^2.$$

MAIN FEATURES

$$\tilde{L} = -\partial_x^2 - \frac{2}{3}\tilde{Q}^3, \quad \tilde{L} < L, \quad \tilde{L}\tilde{H} = 0.$$

Let us consider a perturbation in (2) over $\tilde{\mathbf{H}}$ of the form $\phi = \tilde{\mathbf{H}} + \mathbf{w}$. Then $\mathbf{w}$ satisfies the following system:

$$\begin{cases} \partial_t w_1 = w_2 \\ \partial_t w_2 = -Lw_1 - \tilde{Q}^2(3\tilde{H}w_1^2 + w_1^3), \end{cases}$$

where we have defined the linear operator

$$L\phi = -\partial_x^2\phi + V(x)\phi, \quad \text{with} \quad V(x) = 2\tilde{Q}^2(x)(1 - \tilde{Q}(x)),$$

- ▶ Kink only presenting a polynomial tail.
- ▶ Smallness of a weighted L^∞ norm of the perturbation.
- ▶ Nonlinear terms are as large as the linear ones.
- ▶ Degenerate energy: E has no natural coercivity structure as in standard kink problems.
- ▶ Existence of a weak L^2 threshold resonance associated with $\tilde{L}$.
- ▶ The spectral theory of variable coefficients operators cannot be taken for granted.

$$L = -\partial_x^2 + 2\tilde{Q}^2(1 - \tilde{Q})$$

The operator L satisfies the following properties.

1. The essential spectrum of L is $[0, +\infty)$.
2. The operator L has a unique simple eigenvalue $\lambda_0 = -\mu_0^2$, with associated eigenfunction ϕ_0 that satisfies

$$L\phi_0 = -\mu_0^2\phi_0, \quad |\partial_x^k \phi_0(x)| \lesssim e^{-\frac{\sqrt{2}}{2}\mu_0 x}, \quad k = 0, 1, 2.$$

3. The negative eigenvalue introduce exponentially stable and unstable modes in the neighborhood of the kink.

Remark: The eigenfunction ϕ_0 is not explicit. Nevertheless, we obtain key properties of

$$h_0(x) = \frac{\phi_0'(x)}{\phi_0(x)}.$$

This function is deeply related to the Darboux transformation applied in the dual and its repulsive potential. Also, we have that

$$0.808 \leq \mu_0 \leq 0.883.$$

PREVIOUS WORKS

Kinks in scalar field theories.

- ▶ [Henry-Perez-Wreszinski \(1982\)](#) Orbital stability in the energy space.
- ▶ [Kowalczyk-Martel-Muñoz \(2017\)](#) AS of ϕ^4 kinks under odd data.
- ▶ [Delort and Masmoudi \(2022\)](#) Detailed AS for odd perturbations of the ϕ^4 kink up to times $O(\varepsilon^{-4})$.
- ▶ [Kowalczyk-Martel-Muñoz-Van Den Bosch \(2020\)](#), sufficient condition for general AS of kinks in the energy space.
- ▶ [Kowalczyk-Martel \(2023\)](#): Kink dynamics under odd perturbations with one internal mode (general case: [Cuccagna and Maeda](#)).
- ▶ [Lührmann-Schlag \(2023\)](#): AS of the sine-Gordon (SG) kink under odd perturbations (including resonance).
- ▶ [Maulén-Muñoz \(2024\)](#) Asymptotic stability of the fourth order ϕ^4 kink.

Kinks in variable medium scalar field theories.

1. [Snelson \(2018\)](#). Asymptotic stability for odd perturbations of the stationary kink in the variable-speed ϕ^4 model.
2. [Alammari-Snelson \(2022\)](#) Linear and orbital stability analysis for solitary-wave solutions of variable-coefficient scalar field equations.

EXISTENCE OF A STABLE MANIFOLD

Let $\delta_0 > 0$, we define

$$\mathbf{Y}_{\pm} = \begin{pmatrix} \phi_0 \\ \pm \mu_0 \phi_0 \end{pmatrix}, \quad \mathbf{Z}_{\pm} = \begin{pmatrix} \phi_0 \\ \pm \mu_0^{-1} \phi_0 \end{pmatrix},$$

$$\mathcal{A}_0 = \{ \varepsilon \in H_0(\mathbb{R}) \times L^2(\mathbb{R}) \text{ such that } \|\varepsilon\|_{(H_0 \times L^2)(\mathbb{R})} < \delta_0 \text{ and } \langle \varepsilon, \mathbf{Z}_+ \rangle = 0 \}.$$

Decompose the solution as

$$\begin{cases} \phi(t, x) - \tilde{H} = a_1(t)\phi_0(x) + u_1(t, x) \\ \partial_t \phi(t, x) = \mu_0 a_2(t)\phi_0(x) + u_2(t, x) \end{cases} \quad \text{with } \langle u_1(t), \phi_0 \rangle = 0 = \langle u_2(t), \phi_0 \rangle.$$

Additionally, we set the variables

$$b_+ = \frac{1}{2}(a_1 + a_2), \quad b_- = \frac{1}{2}(a_1 - a_2).$$

Under $\langle u_1, \phi_0 \rangle = 0$, one has

$$\langle Lu_1, u_1 \rangle \gtrsim \|u\|_{H_0(\mathbb{R})}^2.$$

EXISTENCE OF A STABLE MANIFOLD

Theorem (stable manifold)

There exist $C, \delta_0 > 0$ and a Lipschitz function $h : \mathcal{A}_0 \rightarrow \mathbb{R}$ with $h(0) = 0$ and $|h(\varepsilon)| \leq C \|\varepsilon\|_{H_0 \times L^2}^{3/2}$, such that denoting

$$\mathcal{M} = \{\tilde{\mathbf{H}} + \varepsilon + h(\varepsilon)\mathbf{Y}_+ \text{ with } \varepsilon \in \mathcal{A}_0\}, \quad (3)$$

if $(\phi, \partial_t \phi)(0) \in \mathcal{M}$ then the solution $(\phi, \partial_t \phi)$ of (2) with initial data $(\phi, \partial_t \phi)(0)$ is global and satisfies, for all $t \geq 0$,

$$\|\phi(t) - \tilde{\mathbf{H}}\|_{(H_0 \times L^2)(\mathbb{R})} \leq C \|\phi(0) - \tilde{\mathbf{H}}\|_{(H_0 \times L^2)(\mathbb{R})}. \quad (4)$$

$$\tilde{L} = -\partial_x^2 - \frac{2}{3}\tilde{Q}^3, \quad \tilde{L}\tilde{H} = 0.$$

The proof is standard, with a main difference given by **the control of a resonance**. If we expand the energy around the kink,

$$\begin{aligned} & 2 \left\{ E(\phi_1, \phi_2) - E(\tilde{H}, 0) \right\} \\ &= \mu_0^2(a_2^2 - a_1^2) + \|u_2\|_{L^2}^2 + \langle Lu_1, u_1 \rangle + \frac{1}{2} \int \tilde{Q}^2(u_1^2 + 4\tilde{H}u_1)u_1^2 \\ &\quad + 2a_1 \int \tilde{Q}^2\phi_0u_1^3 + 3a_1 \int \tilde{Q}^2(2\tilde{H} + a_1\phi_0)\phi_0u_1^2 + 2a_1^2 \int \tilde{Q}^2(3\tilde{H} + a_1\phi_0)\phi_0^2u_1 \\ &\quad + \frac{1}{2}a_1^3 \int \tilde{Q}^2(4\tilde{H} + a_1\phi_0)\phi_0^3 \\ &=: l_1 + \underbrace{l_2 + l_3}_{\text{controlled}} \lesssim |a_1|^3 + \varepsilon \|\tilde{Q}u_1\|_{L^2}^2. \end{aligned}$$

The term l_1 can be written as

$$l_1 = \mu_0^2(a_2^2 - a_1^2) + \|u_2\|_{L^2}^2 + \langle \tilde{L}u_1, u_1 \rangle + \frac{1}{2} \int \tilde{Q}^2(u_1 + 2\tilde{H})^2u_1^2.$$

PROOF EXISTENCE OF A STABLE MANIFOLD

This implies

$$\begin{aligned} -4\mu_0^2 b_+ b_- + \|u_2\|_{L^2}^2 + \langle \tilde{L}u_1, u_1 \rangle + \int \tilde{Q}^2(u_1 + 2\tilde{H})^2 u_1^2 \\ \leq C\{E(\phi_1, \phi_2) - E(\tilde{H}, 0)\} + C|a_1|^3 + \varepsilon \|u_1\|_{H_0}^2. \end{aligned}$$

We decompose u_1 now as

$$u_1 = \mathbf{a}(t)\tilde{H} + \tilde{u}_1, \quad \text{with} \quad \langle \phi_0, \tilde{u}_1 \rangle = \langle \tilde{Q}^2 \tilde{H}^3, \tilde{u}_1 \rangle = 0.$$

Note that $a(t)$ is well-defined, $\langle \tilde{L}u_1, u_1 \rangle = \langle \tilde{L}\tilde{u}_1, \tilde{u}_1 \rangle$, and we have

$$a^2 \|\tilde{H}\|_{H_0}^2 + \|\tilde{u}_1\|_{H_0}^2 = \|u_1\|_{H_0}^2.$$

Let $\delta_0 > 0$ be defined by

$$\delta_0^2 = b_+^2(0) + b_-^2(0) + \|u_2(0)\|_{L^2}^2 + \|u_1(0)\|_{H_0}^2 + \left\| \tilde{Q}(u_1(0) + 2\tilde{H})u_1(0) \right\|_{L^2}^2.$$

Then $|E(\phi_1, \phi_2) - E(\tilde{H}, 0)| \lesssim \delta_0^2$.

ESTIMATE FOR BOOTSTRAP ARGUMENT

Thus

$$\|u_2\|_2^2 + \langle \tilde{L}\tilde{u}_1, \tilde{u}_1 \rangle + \int \tilde{Q}^2(u_1 + 2\tilde{H})^2 u_1^2 \lesssim \delta_0^2 + b_+^2 + b_-^2 + |a_1|^3 + \varepsilon \|u_1\|_{H_0}^2.$$

For the NL term:

$$\int \tilde{Q}^2(u_1 + 2\tilde{H})^2 u_1^2 \gtrsim a^2(2+a)^2 - |a| \|u_1\|_{H_0}^2.$$

For the linear operator, we prove the coercivity

$$\langle \tilde{L}\tilde{u}_1, \tilde{u}_1 \rangle \gtrsim \|\tilde{u}_1\|_{H_0}^2.$$

Then, taking δ_0 and ε small, we have

$$\boxed{a^2 + \|u_2\|_{L^2}^2 + \|\tilde{u}_1\|_{H_0}^2 \lesssim |b_+|^2 + |b_-|^2 + \delta_0^2 + O(|b_+|^3, |b_-|^3, \|u_1\|_{H_0}^3)} \quad (5)$$

This estimate allows us to conclude using a classical bootstrap argument.

Take $\varepsilon = (\varepsilon_1, \varepsilon_2) \in \mathcal{A}_0$, and decompose the initial data as

$$(\phi, \partial_t \phi)(0) = (1 + a(0))\tilde{\mathbf{H}} + (\tilde{u}_1, u_2)(0) + b_-(0)\mathbf{Y}_- + h(\varepsilon)\mathbf{Y}_+.$$

We will prove there exists $h(\varepsilon) := b_+(0)$ such that the solution is global and satisfies (4).

For $\delta_0 > 0$ small, $K > 1$ large, we introduce

$$|a| \leq K^2 \delta_0, \|\tilde{u}_1\|_{H_0} \leq K^2 \delta_0, \|u_2\|_{L^2} \leq K^2 \delta_0 \quad \text{and} \quad |b_-| \leq K \delta_0, \quad (6)$$

$$|b_+| \leq K^5 \delta_0^2. \quad (7)$$

We consider

$$\begin{aligned} |a(0)| &\leq \delta_0, \|\tilde{u}_1(0)\|_{H_0} \leq \delta_0, \|\tilde{Q}(u_1(0) + 2\tilde{H})u_1(0)\|_{L^2} \leq \delta_0, \\ \|u_2(0)\|_{L^2} &\leq \delta_0, |b_-(0)| \leq \delta_0, |b_+(0)| \leq K^5 \delta_0. \end{aligned}$$

BOOTSTRAP

Let

$$T = \sup \{t \geq 0 \text{ such that (6), (7), hold on } [0, t]\},$$

well-defined since $K > 1$.

We will prove that there exists at least a value of $b_+(0) \in [-K^5 \delta_0^2, K^5 \delta_0^2]$ such that $T = \infty$. By contradiction, assuming that any $b_+(0) \in [-K^5 \delta_0^2, K^5 \delta_0^2]$ leads to $T < \infty$.

- ▶ If $T < +\infty$, from (6) and (7)

$$a^2 + \|\tilde{u}_1\|_{H_0}^2 + \|u_2\|_{L^2}^2 \leq 3K^4 \delta_0^2.$$

- ▶ But using (5) for K large fixed, and δ_0 small, we get

$$a^2 + \|\tilde{u}_1\|_{H_0}^2 + \|u_2\|_{L^2}^2 \leq \frac{3}{4} K^4 \delta_0^2$$

and similarly for b_-^2 .

- ▶ The above implies that $b_+(T) = K^5 \delta_0^2$, and allows us to conclude.

Theorem (asymptotic stability)

There exists $\delta > 0$ such that if a global solution $\phi \in \mathbf{E}$ of (2) satisfies

$$\sup_{t \geq 0} \left\| \phi(t) - \tilde{\mathbf{H}} \right\|_{(H_0 \times L^2)(\mathbb{R})} < \delta, \quad (8)$$

then for any I bounded interval in $\mathbb{R}$,

$$\lim_{t \rightarrow \infty} \left\| \phi(t) - \tilde{\mathbf{H}} \right\|_{(H^1 \times L^2)(I)} = 0. \quad (9)$$

One can easily show that global existence with small bounds imply

$$\|u_1(t)\|_{H_0} + \|u_2(t)\|_{L^2} + |a_1(t)| + |a_2(t)| + |b_+(t)| + |b_-(t)| \lesssim \delta.$$

L^∞ smallness: One has $\|\tilde{Q}^{\frac{1}{2}} u\|_{L^\infty(\mathbb{R})} \lesssim \|u\|_{H_0(\mathbb{R})}$.

PROOF OF AS

We have that (b_-, b_+) satisfies the system

$$\text{Unstable modes} \quad \begin{cases} \dot{b}_+(t) = \mu_0 b_+(t) - \frac{N_0}{2\mu_0} \\ \dot{b}_-(t) = -\mu_0 b_-(t) + \frac{N_0}{2\mu_0}, \end{cases}$$

where

$$N = \tilde{Q}^2 \left(3\tilde{H}(a_1\phi_0 + u_1)^2 + (a_1\phi_0 + u_1)^3 \right),$$

$$N^\perp = N - N_0\phi_0, \quad N_0 = \langle N, \phi_0 \rangle,$$

and (u_1, u_2) satisfies the following system

$$\text{Radiating states} \quad \begin{cases} \dot{u}_1 = u_2 \\ \dot{u}_2 = -Lu_1 - N^\perp. \end{cases}$$

LOCAL WEIGHTS AND SCALES

$$\zeta_A^2(x) = \exp\left(-\frac{1}{A}|\alpha^{-1}(x)|(1-\chi(x))\right).$$

$$\varphi_A(x) = \int_0^x \tilde{\alpha} \zeta_A^2(y) dy$$

Let $\chi \in C_c^\infty(\mathbb{R})$ be a smooth even function satisfying

$$\chi(x) = 1 \text{ for } |x| \leq 1, \quad \chi(x) = 0 \text{ for } |x| \geq 2, \quad \chi'(x) \leq 0 \text{ for } x \geq 0.$$

For $A > 0$, let ζ_A and ϕ_A be weighed functions

$$\zeta_A(x) \approx \begin{cases} 1 & \text{for } |x| \leq 1, \\ \frac{1}{|x|^{\frac{1}{A}}} & \text{for } |x| \geq 1. \end{cases} \quad \tilde{\chi}_A(x) = \chi\left(\frac{\alpha^{-1}(x)}{A}\right).$$

$$\varphi_A(x) \approx \begin{cases} |\alpha^{-1}(x)| & \text{for } |x| \leq 1, \\ O(A) & \text{for } |x| \gg 1. \end{cases}$$

For $B > 0$, we define ζ_B , φ_B , and $\tilde{\chi}_B$ analogously, and

$$\psi_{A,B}(x) = \tilde{\chi}_A^2(x) \varphi_B(x).$$

Assume

$$1 \ll B \ll A.$$

FIRST VIRIAL

$$\begin{cases} \dot{u}_1 = u_2 \\ \dot{u}_2 = -Lu_1 - N^\perp. \end{cases}$$

Let

$$\mathcal{J}(t) = \int \left(\varphi_A \partial_x u_1 + \frac{1}{2} \varphi'_A u_1 \right) u_2, \quad N = \tilde{Q}^2 \left(3\tilde{H}(a_1 \phi_0 + u_1)^2 + (a_1 \phi_0 + u_1)^3 \right).$$

and introduce the variables

$$w_i = \zeta_A u_i, \quad i = 1, 2.$$

Computing, replacing and IBP, we have

$$\frac{d}{dt} \mathcal{J} = - \underbrace{\int \varphi'_A (\partial_x u_1)^2 + \frac{1}{4} \int \varphi_A''' u_1^2 + \frac{1}{2} \int \varphi_A V' u_1^2}_{I_L} - \underbrace{\int \left(\varphi_A \partial_x u_1 + \frac{1}{2} \varphi'_A u_1 \right) N^\perp}_{I_N}.$$

After (tedious) computations using estimates on φ_A , N , N_0 , the decay of ϕ_0 , $\tilde{Q}$ and completion of squares, we obtain

$$\begin{aligned} -I_N &\leq Ca_1^4 + C \int \tilde{Q}^7 u_1^2 + CA \|\tilde{Q}^{1/2} u_1\|_{L^\infty} \int \tilde{Q}^3 w_1^2 \quad \longrightarrow \text{"controlled"} \\ &\quad + \frac{1}{72} \int \tilde{Q}^3 \tilde{H}^2 w_1^2 + \frac{4}{9} \int \tilde{Q}^3 \left| \varphi_A \tilde{H} \right| \tilde{H}^2 u_1^2. \quad =: \tilde{I}_N \text{ positive quadratic} \end{aligned}$$

Rewriting using the variables (w_1, w_2) ,

$$I_L + \tilde{I}_N = - \int \tilde{Q} (\partial_x w_1)^2 + \frac{1}{2} \int \left[\frac{\zeta_A''}{\zeta_A} - \left(\frac{\zeta_A'}{\zeta_A} \right)^2 \right] \tilde{Q} w_1^2 + \int V_A u_1^2,$$

where

$$V_A(x) := \frac{1}{4} \left(\tilde{Q}'' + \frac{1}{18} \tilde{Q}^3 \tilde{H}^2 \right) \zeta_A^2 + \frac{1}{2} \varphi_A V' + \frac{4}{9} \left| \varphi_A \tilde{H} \right| \tilde{H}^2 \tilde{Q}^3.$$

The colored terms must compete each other via the following estimates:

$$\left| \frac{\zeta_A''}{\zeta_A} - \left(\frac{\zeta_A'}{\zeta_A} \right)^2 \right| \tilde{Q} \lesssim \frac{1}{A} \tilde{Q}^3 \mathbf{1}_{\{|x| \geq 1\}}, \quad V_A(x) \mathbf{1}_{\{|x| \geq \tilde{x}\}} \leq -C \tilde{Q}^3(x).$$

With this, and taking A large, we can conclude that

$$\begin{aligned} I_L + \tilde{I}_N &\leq - \int \tilde{Q} (\partial_x w_1)^2 + \int_{|x| \gtrsim 1} \left(\frac{C'}{A} - C \right) \tilde{Q}^3 u_1^2 + C \int_{|x| \lesssim 1} \tilde{Q}^2 u_1^2 \\ &\leq -C_0 \int \tilde{Q} [(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + C \int \tilde{Q}^7 u_1^2. \end{aligned}$$

Thus,

$$\begin{aligned} \frac{d}{dt} \mathcal{J} \leq & -C_0 \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + C a_1^4 + C \int \tilde{Q}^7 u_1^2 \\ & + C A \|\tilde{Q}^{1/2} u_1\|_{L^\infty} \int \tilde{Q}^3 w_1^2 \end{aligned}$$

and we obtain the following result:

Proposition 1

There exist $C_0, C > 0$ and $\delta_1 > 0$ such that for any $0 < \delta \leq \delta_1$, the following holds. Fix

$$A = \delta^{-1/4}.$$

Under global existence and smallness of the perturbation, for all $t \geq 0$, the functional $\mathcal{J}$ satisfies the estimate

$$\boxed{\frac{d}{dt} \mathcal{J} \leq -\frac{1}{2} C_0 \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + C \int \tilde{Q}^7 u_1^2 + C |a_1|^4.}$$

TRANSFORMED PROBLEM (DARBOUX TRANSFORMATION)

To control the term $\int \tilde{Q}^7 u_1^2$ we employ the *Darboux transformation*

$$L_0 = -\partial_x^2 + V_0, \quad \text{with} \quad V_0 := 2(h_0^2 - \mu_0^2) - V,$$

$$U = \phi_0 \cdot \partial_x \cdot \phi_0^{-1}, \quad U^* = -\phi_0^{-1} \cdot \partial_x \cdot \phi_0.$$

1. We have $L = U^* U - \mu_0^2$, $L_0 = U U^* - \mu_0^2$ and $U L = L_0 U$.
2. Set $v_1 = U u_1$, $v_2 = U u_2$. Then the linear part of the transformed system is

$$\begin{cases} \dot{v}_1 = v_2 \\ \dot{v}_2 = -L_0 v_1. \end{cases}$$

3. Despite the fact that ϕ_0 is not explicit, we achieved to prove that V_0 **is positive and repulsive** (Reed-Simon IV, Theorem XIII.60). In particular, for $x \geq 0$ we obtain the integral formula

$$V'_0(x) = \frac{h_0}{\phi_0^2} \int_x^\infty I(y) \left(\frac{\phi_0}{h_0} \right)^2(y) dy, \quad \text{with} \quad I(y) \geq 0.$$

TRANSFORMED SYSTEM

We define $X_\gamma : L^2(\mathbb{R}) \rightarrow H^2(\mathbb{R})$, $X_\gamma = (1 - \gamma \partial_x^2)^{-1}$ and set

$$\begin{cases} v_1 = X_\gamma U(\tilde{\chi}_B u_1), \\ v_2 = X_\gamma U(\tilde{\chi}_B u_2), \end{cases}$$

With this we have $(v_1, v_2) \in (H_0 \cap \dot{H}^2)(\mathbb{R}) \times H^1(\mathbb{R})$ and the system

$$\begin{cases} \dot{v}_1 = v_2 \\ \dot{v}_2 = -L_0 v_1 - \gamma X_\gamma (V_0'' v_1 + 2V_0' \partial_x v_1) \\ \quad - X_\gamma U[2\tilde{\chi}_B' \partial_x u_1 + \tilde{\chi}_B'' u_1] - X_\gamma U(\tilde{\chi}_B N^\perp). \end{cases} \quad (10)$$

SECOND VIRIAL ESTIMATE

Let

$$\mathcal{J}(t) = \int \left(\psi_{A,B}(x) \partial_x v_1(t, x) + \frac{1}{2} \psi'_{A,B}(x) v_1(t, x) \right) v_2(t, x) dx$$

with $\psi_{A,B} = \tilde{\chi}_A^2 \phi_B$, and

$$z := \tilde{\chi}_A \zeta_B v_1.$$

Analogously to the first virial, computing and IBP we obtain

$$\begin{aligned} \frac{d}{dt} \mathcal{J} &= \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) \dot{v}_2 \\ &= - \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) L_0 v_1 \\ &\quad - \gamma \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) X_\gamma (V_0'' v_1 + 2 V_0' \partial_x v_1) \\ &\quad - \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) X_\gamma U [2 \tilde{\chi}_B' \partial_x u_1 + \tilde{\chi}_B'' u_1] \\ &\quad - \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) X_\gamma U (\tilde{\chi}_B N^\perp) \\ &=: J_1 + J_2 + J_3 + J_4. \end{aligned}$$

CONTROL FOR J_1

We have

$$J_1 = - \int \psi'_{A,B} (\partial_x v_1)^2 + \frac{1}{4} \int \psi'''_{A,B} v_1^2 - \int \left(\psi_{A,B} \partial_x v_1 + \frac{1}{2} \psi'_{A,B} v_1 \right) v_0 v_1.$$

Using the definitions of $\psi_{A,B}$ and introducing z we get

$$J_1 = - \int [\tilde{Q} (\partial_x z)^2 + \textcolor{red}{V}_B z^2] + \tilde{J}_1,$$

with

$$V_B = \left[\frac{1}{2} \tilde{Q} \left(\frac{\zeta_B''}{\zeta_B} - \frac{(\zeta_B')^2}{\zeta_B^2} \right) - \frac{1}{10} \frac{\varphi_B}{\zeta_B^2} V_0' \right] + \left[-\frac{1}{4} \tilde{Q}'' - \frac{2}{5} \frac{\varphi_B}{\zeta_B^2} V_0' \right] =: \textcolor{red}{V}_B^I + \textcolor{red}{V}_B^{II} \geq 0$$

New goal!

CONTROL FOR J_1

- ▶ For V_B^I we have the following control for B large

$$V_B^I \geq \underbrace{-\frac{C}{B} \tilde{Q}^3 \mathbf{1}_{\{|x| \geq 1\}}}_{\text{neg. but small}} + \underbrace{\frac{1}{10} \alpha^{-1}(x) |V_0'|}_{\gtrsim \tilde{Q}^3} \gtrsim \tilde{Q}^3 \mathbf{1}_{\{|x| \geq 1\}}.$$

- ▶ For V_B^{II} we do not have smallness w.r.t B , but **employing the accurate estimates of V_0'**

$$3V'(x) \leq V_0'(x) \leq \frac{1}{2}V'(x),$$

for $x \gg 1$, and the identity

$$-\frac{1}{4}\tilde{Q}'' = \frac{1}{2}\tilde{Q}^3 \left(\frac{5}{6}\tilde{Q} - 1 \right),$$

we succeeded on proving that

$$V_B^{II} \gtrsim \tilde{Q}^3.$$

CONTROL OF ERROR TERMS

Now we have the estimate

$$\frac{d}{dt} \mathcal{J} \leq - \int \tilde{Q} \left[(\partial_x z)^2 + c_1 \tilde{Q}^2 z^2 \right] + \underbrace{\tilde{J}_1 + J_2 + J_3 + J_4}_{\text{technical error \& NL terms}},$$

Using some weighted estimates relating the variables v, w and z , for the error terms we have that

$$|\tilde{J}_1| \lesssim \gamma^{-1} \frac{B}{A} \left(\int \tilde{Q} (\partial_x w_1)^2 + \int \tilde{Q}^3 w_1^2 \right).$$

$$|J_2| \lesssim \gamma \left(\int \tilde{Q} (\partial_x z)^2 + \int \tilde{Q}^3 z^2 \right) + e^{-\frac{A}{4}} \left(\int \tilde{Q} (\partial_x w_1)^2 + \int \tilde{Q}^3 w_1^2 \right).$$

$$|J_3| \lesssim \gamma^{-1} B^{-1} \left(\left\| \tilde{Q}^{\frac{1}{2}} \partial_x w_1 \right\| + \left\| \tilde{Q}^{\frac{3}{2}} w_1 \right\| \right)^2 + \gamma^{-\frac{1}{2}} B^{-2} \left(\left\| \tilde{Q}^{\frac{1}{2}} \partial_x w_1 \right\| + \left\| \tilde{Q}^{\frac{3}{2}} w_1 \right\| \right) \left\| \tilde{Q}^{3/2} z \right\|.$$

For J_4 we decompose $N = N_g + N_b$ where

$$N_g := \tilde{Q}^2 (3\tilde{H} (a_1^2 \phi_0^2 + 2a_1 \phi_0 u_1) + \dots),$$

Small due to a_1, ϕ_0
 L^∞ control

$$N_b := \tilde{Q}^2 u_1^2 (3\tilde{H} + u_1).$$

Problem with L^∞ control
Use localization from $\tilde{x}_B$

CONTROL OF ERROR TERMS

We obtain

$$|J_4| \lesssim \gamma^{-1} B \left(\left\| \tilde{Q}^{\frac{1}{2}} \partial_x w_1 \right\| + \left\| \tilde{Q}^{\frac{3}{2}} w_1 \right\| \right) K_B(a_1, u_1, w_1) + \gamma^{-\frac{1}{2}} \left\| \tilde{Q}^{\frac{3}{2}} z \right\| K_B(a_1, u_1, w_1).$$

where

$$K_B(a_1, u_1, w_1) = a_1^2 + \alpha(B) \left(1 + \alpha(B) \|\tilde{Q}^{1/2} u_1\|_{L^\infty} \right) \|\tilde{Q}^{1/2} u_1\|_{L^\infty} \|\tilde{Q}^{\frac{3}{2}} w_1\|.$$

Using these estimates in the virial inequality, and taking $\gamma > 0$ small but fixed such that the first term controls the variable z , we have

$$\begin{aligned} \frac{d}{dt} \mathcal{J} \leq & -c_2 \int \tilde{Q} [(\partial_x z)^2 + \tilde{Q}^2 z^2] + C \left(\frac{1}{B} + \frac{B}{A} + e^{-\frac{A}{4}} \right) \int \tilde{Q} [(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] \\ & + CB\alpha(B)K_B(a_1, u_1, w_1)^2. \end{aligned}$$

Proposition 2

There exist $C_2, C > 0$ and $\delta_2 > 0$ such that for γ small enough and for any $0 < \delta \leq \delta_2$, the following holds. Fix

$$B = \alpha^{-1}(\delta^{-1/8}) \approx \ln(\delta^{-\frac{1}{8}}),$$

and assume that for all $t \geq 0$, the smallness bound holds. Then, for all $t \geq 0$, $\mathcal{J}$ satisfies

$$\begin{aligned} \frac{d}{dt} \mathcal{J} \leq & -C_2 \int \tilde{Q}[(\partial_x z)^2 + \tilde{Q}^2 z^2] \\ & + C \ln(\delta^{-\frac{1}{8}})^{-1} \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + C \delta^{\frac{1}{2}} |a_1|^3. \end{aligned}$$

In addition, we proved the following transfer estimate:

$$\int \tilde{Q}^7 u_1^2 \lesssim \int \tilde{Q}^2[(\partial_x z)^2 + \tilde{Q}^2 z^2] + e^{-\frac{A}{4}} \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2]. \quad (11)$$

END OF PROOF MAIN THEOREM

If we define for $0 < \delta < \delta_3$,

$$\mathcal{H} = \mathcal{J} + 8CC_0^{-1} \ln(\delta_3^{-\frac{1}{8}})^{-1} \mathcal{I},$$

using Proposition 1 and 2, and the transfer estimate (11), then we have

$$\begin{aligned} \frac{d}{dt} \mathcal{H} &\leq -C_2 \int \tilde{Q}[(\partial_x z)^2 + \tilde{Q}^2 z^2] + C \ln(\delta^{-\frac{1}{8}})^{-1} \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] \\ &\quad + C \ln(\delta_3^{-\frac{1}{8}})^{-1} \int \tilde{Q}[(\partial_x z)^2 + \tilde{Q}^2 z^2] \\ &\quad - 2C \ln(\delta_3^{-\frac{1}{8}})^{-1} \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + |a_1|^3. \\ &\leq -C \ln(\delta_3^{-\frac{1}{8}})^{-1} \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + |a_1|^3. \end{aligned}$$

Then, for all $t \geq 0$,

$$\frac{d}{dt} \mathcal{H} \leq -C_3 \int \tilde{Q} [(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] + |a_1|^3.$$

END OF PROOF

We define now

$$\mathcal{B} = b_+^2 - b_-^2.$$

Lemma

There exist $C_4 > 0$ and $0 < \delta_4 \leq \delta_3$ such that for any $0 < \delta \leq \delta_4$, the following holds. Then,

$$\frac{d}{dt} \mathcal{B} \geq \mu_0(b_+^2 + b_-^2) - C_4 \int \tilde{Q}^3 w_1^2.$$

Combining these previous estimates, for possibly smaller $\delta > 0$,

$$\begin{aligned} \frac{d}{dt} (\mathcal{B} - 2C_4 C_3^{-1} \mathcal{H}) &\geq \mu_0(b_+^2 + b_-^2) - C_4 \int \tilde{Q}^3 w_1^2 + 2C_4 \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] \\ &\quad - 2C_4 C_3^{-1} |a_1|^3 \\ &\geq \frac{\mu_0}{8} (a_1^2 + a_2^2) + C_4 \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] \end{aligned}$$

We have

$$a_1^2 + a_2^2 + \int \tilde{Q}[(\partial_x w_1)^2 + \tilde{Q}^2 w_1^2] \lesssim \frac{d}{dt} (\mathcal{B} - 2C_4 C_3^{-1} \mathcal{H})$$

From the smallness of the bounds and the choice of A, B that $|\mathcal{I}|, |\mathcal{J}| \lesssim \delta$. Then, we have

$$|\mathcal{H}| \lesssim \delta, \quad |\mathcal{B}| \lesssim \delta^2.$$

Therefore, integrating the equation above on $[0, t]$, passing to the limit as $t \rightarrow +\infty$, and recovering u_1 and u_2 , it follows that

$$\int_0^\infty \left\{ a_1^2 + a_2^2 + \underbrace{\int \tilde{Q}^2 [(\partial_x u_1)^2 + \tilde{Q}^{\frac{3}{2}} u_1^2 + u_2^2]}_{=: \mathcal{G}} \right\} dt \lesssim \delta.$$

We deduce

$$\int_0^\infty [a_1^2 + a_2^2 + \mathcal{G}] dt \lesssim \delta.$$

From the previous slide, there exists an increasing sequence $t_n \rightarrow +\infty$ such that

$$\lim_{n \rightarrow \infty} [a_1^2(t_n) + a_2^2(t_n) + \mathcal{G}(t_n)] = 0.$$

In addition, we obtain for $\mathcal{G}$ that

$$\left| \frac{d}{dt} \mathcal{G} \right| \lesssim a_1^2 + \mathcal{G}.$$

For $t \geq 0$, integrating on $[t, t_n]$, and passing to the limit as $n \rightarrow \infty$, we obtain

$$\mathcal{G}(t) \lesssim \int_t^\infty [a_1^2 + \mathcal{G}] dt.$$

Using the integrability of a_1 and $\mathcal{G}$, we deduce that $\lim_{t \rightarrow \infty} \mathcal{G}(t) = 0$.

One can verify that

$$\left| \frac{d}{dt}(a_1^2) \right| + \left| \frac{d}{dt}(a_2^2) \right| \lesssim a_1^2 + a_2^2 + \int \tilde{Q}^{\frac{7}{2}} u_1^2.$$

Similarly as before, by integration on $[t, t_n]$ and taking $n \rightarrow \infty$,

$$a_1^2(t) + a_2^2(t) \lesssim \int_t^\infty [a_1^2 + a_2^2 + \mathcal{G}] dt,$$

which proves $\lim_{t \rightarrow \infty} (|a_1(t)| + |a_2(t)|) = 0$.

By the decomposition of the solution, this complete the proof of the main theorem.

SUMMARY OF PROOF

- ▶ We have proved that if $(\phi, \partial_t \phi)$ solves (2), orbitally stable, and decomposed as

$$\begin{cases} \phi(t, x) = \tilde{H} + a_1(t)\phi_0(x) + u_1(t, x) \\ \partial_t \phi(t, x) = \mu_0 a_2(t)\phi_0(x) + u_2(t, x) \end{cases}$$

with $\langle u_1(t), \phi_0 \rangle = 0 = \langle u_2(t), \phi_0 \rangle$, then

$$\lim_{t \rightarrow \infty} |a_1(t)| + |a_2(t)| + \|(u_1, u_2)\|_{(H^1_{loc} \times L^2_{loc})} = 0.$$

- ▶ This proves the asymptotic stability of the kink $\tilde{H}$.

Thanks for your attention !